

Document Name:	Test_Case_Report.doc	Drafter:	Muhammad Fizry Bin Ridzwan
Document Version:	1.0	Tester:	Muhammad Fizry Bin Ridzwan
Python Version:	3.10.11	GitHub Link:	https://github.com/FrostTechHub/Image_Processing_Pipeline

Test Case ID	Scenario	Input	Expected Output	Actual Output	Status
Upload & Validation					
TC_001	Upload a valid JPG file	curl.exe -F "file=@cat_01.jpg" http://127.0.0.1:8000/api/images	{ "status": "queued", "data": { "image_id": "<image_id>", "original_name": "<file_name> ", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null, "error": null } }	{ "status": "queued", "data": { "image_id": "img671", "original_name": "cat_01.jpg", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null, "error": null } }	Pass
Evidence: <pre> PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe -F "file=@cat_01.jpg" http://127.0.0.1:8000/api/images {"status": "queued", "data": {"image_id": "img671", "original_name": "cat_01.jpg", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null}, "error": null} PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> </pre>					

TC_002	Upload a valid PNG file	url.exe -F "file=@cat_03.png" http://127.0.0.1:8000/api/images	{ "status": "queued", "data": { "image_id": "<image_id>", "original_name": "<file_name> ", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null, "error": null }	{ "status": "queued", "data": { "image_id": "img135", "original_name": "cat_03.png", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null, "error": null }	Pass
Evidence: <pre>PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe -F "file=@cat_03.png" http://127.0.0.1:8000/api/images {"status": "queued", "data": {"image_id": "img135", "original_name": "cat_03.png", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null}, "error": null}</pre>					
TC_003	Upload unsupported file types	curl.exe -F "file=@book_1.xlsx" http://127.0.0.1:8000/api/images	{ "status": "failed", "data": { "image_id": "<image_id>", "original_name": "<file_name> ", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null, "error": "<error message> " }	{ "status": "failed", "data": { "image_id": "img029", "original_name": "book_1.xlsx", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null, "error": "Invalid File Format!" }	Pass
Evidence: <pre>PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe -F "file=@book_1.xlsx" http://127.0.0.1:8000/api/images {"status": "failed", "data": {"image_id": "img029", "original_name": "book_1.xlsx", "processed_at": "", "metadata": null, "exif_data": null, "ai_caption": "", "thumbnails": null}, "error": "Invalid File Format!"} PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline></pre>					

TC_004	Upload same filename twice	curl.exe -F "file=@cat_01.jpg" http://127.0.0.1:8000/api/images {"status":"queued", "data":{"image_id":"<image_id>", "original_name":"<file_name> ", "processed_at":"","metadata":null, "exif_data":null, "ai_caption":""," "thumbnails":null}, "error":null} {"status":"queued", "data":{"image_id":"<image_id>", "original_name":"<file_name> ", "processed_at":"","metadata":null, "exif_data":null, "ai_caption":""," "thumbnails":null}, "error":null} *image_id should be different from the one above...	{"status":"queued", "data":{"image_id":"img671", "original_name":"cat_01.jpg", "processed_at":""," "metadata":null, "exif_data":null, "ai_caption":""," "thumbnails":null},"error":null} {"status":"queued", "data":{"image_id":"img935", "original_name":"cat_01.jpg", "processed_at":""," "metadata":null, "exif_data":null, "ai_caption":""," "thumbnails":null},"error":null}	Pass
--------	----------------------------	--	--	------

Evidence:

```

PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe -F "file=@cat_01.jpg" http://127.0.0.1:8000/api/images
{"status":"queued","data":{"image_id":"img671","original_name":"cat_01.jpg","processed_at":"","metadata":null,"exif_data":null,"ai_caption":"","thumbnails":null},"error":null}
PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline>
PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline>
PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe -F "file=@cat_01.jpg" http://127.0.0.1:8000/api/images
{"status":"queued","data":{"image_id":"img932","original_name":"cat_01.jpg","processed_at":"","metadata":null,"exif_data":null,"ai_caption":"","thumbnails":null},"error":null}
PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline>

```

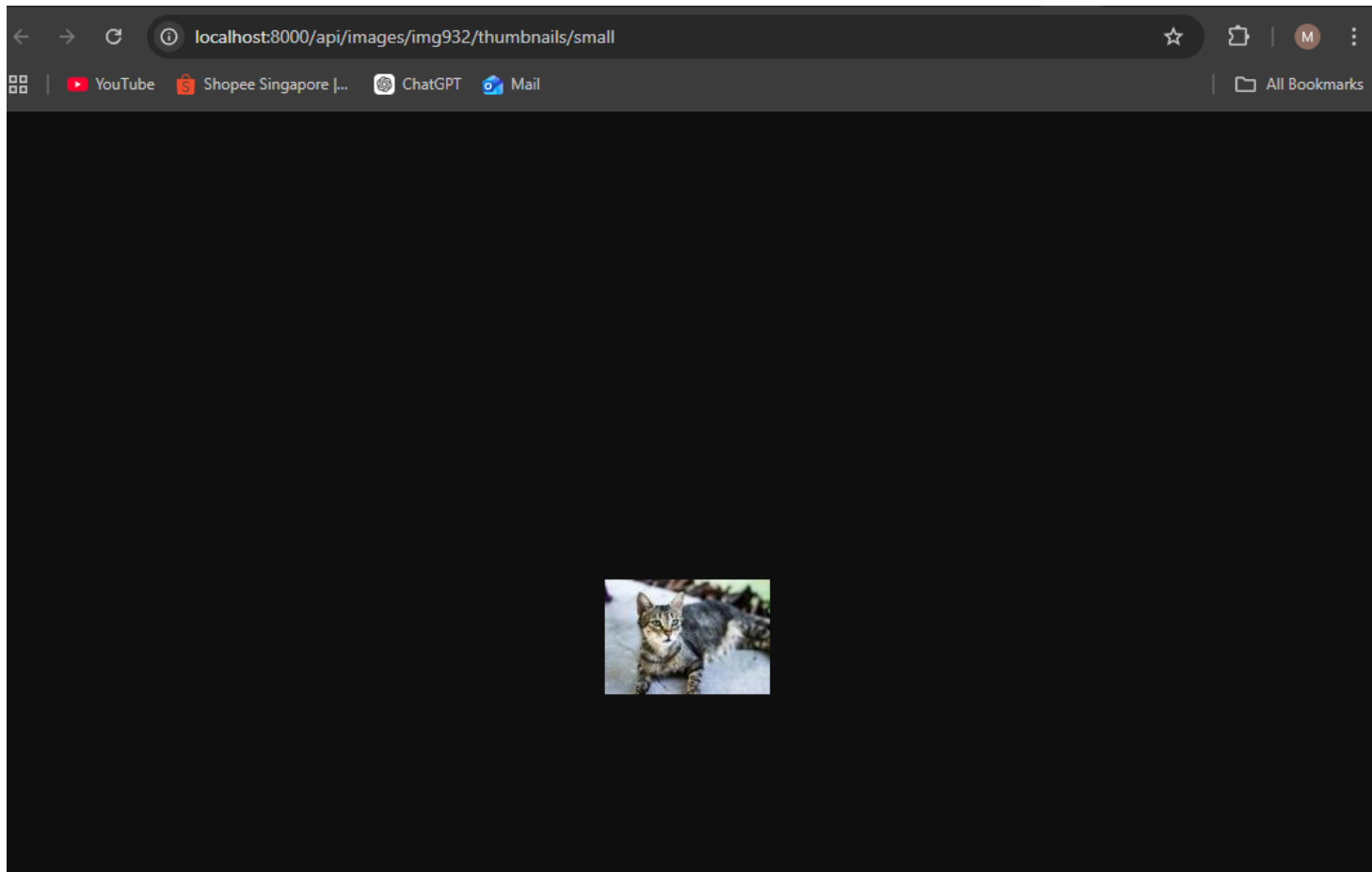
Metadata / Details Retrieval					
TC_005	Get details with valid image_id	curl.exe http://127.0.0.1:8000/api/images/img932	*Success Message	*Refer to Evidence cell	Pass
Evidence: <pre> PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe http://127.0.0.1:8000/api/images/img932 {"status":"success","data":{"image_id":"img932","original_name":"cat_01.jpg","processed_at":"2026-02-23T11:55:09Z","metadata":{"width":270,"height":187,"format":"JPEG","size_bytes":7889},"exif_data":{"info":"No EXIF Data to extract"},"ai_caption":"a cat sitting on the ground next to a wall","thumbnails":{"small":"http://localhost:8000/api/images/img932/thumbnails/small","medium":"http://localhost:8000/api/images/img932/thumbnails/medium"}}, "error":null} PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> </pre>					
TC_006	Get details with invalid / non-existent image_id	curl.exe http://127.0.0.1:8000/api/images/img936	{"status":"failed","data":null,"error":"Invalid Image ID - Image does not exists..."}		Pass
Evidence: <pre> PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe http://127.0.0.1:8000/api/images/img936 {"status":"failed","data":null,"error":"Invalid Image ID - Image does not exists..." } PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> </pre>					

TC_007	Get details with malformed image_id	curl.exe http://127.0.0.1:8000/api/images/im000	{"status":"failed","data":null,"error":"Invalid Image ID - Image does not exists..."}		Pass
--------	-------------------------------------	--	---	--	------

Evidence:

```
PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe http://127.0.0.1:8000/api/
images/img000
{"status":"failed","data":null,"error":"Invalid Image ID - Image does not exists..."}
PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline>
```

Thumbnail Retrieval					
TC_008	Retrieve small thumbnail successfully	*Entered into the browser... http://localhost:8000/api/images/img932/thumbnails/small	Image of cat_01.jpg but in small size...	*Refer to Evidence	Pass
Evidence:					



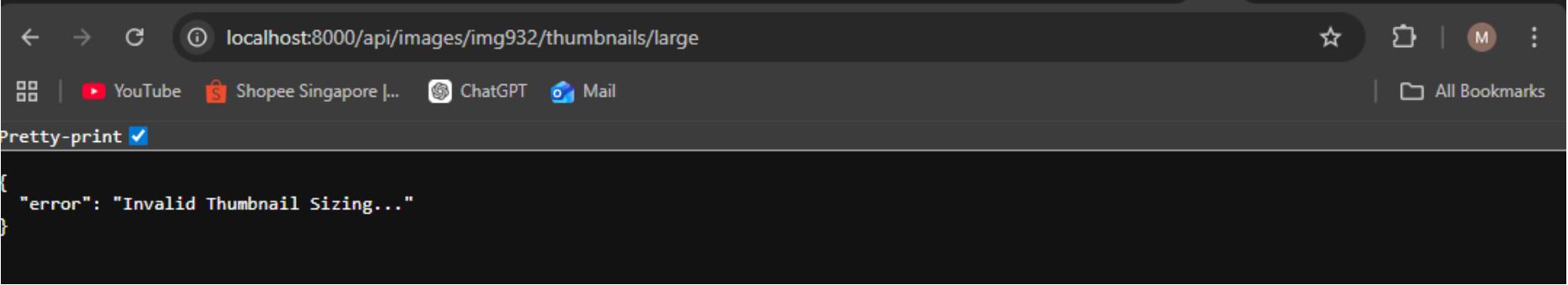
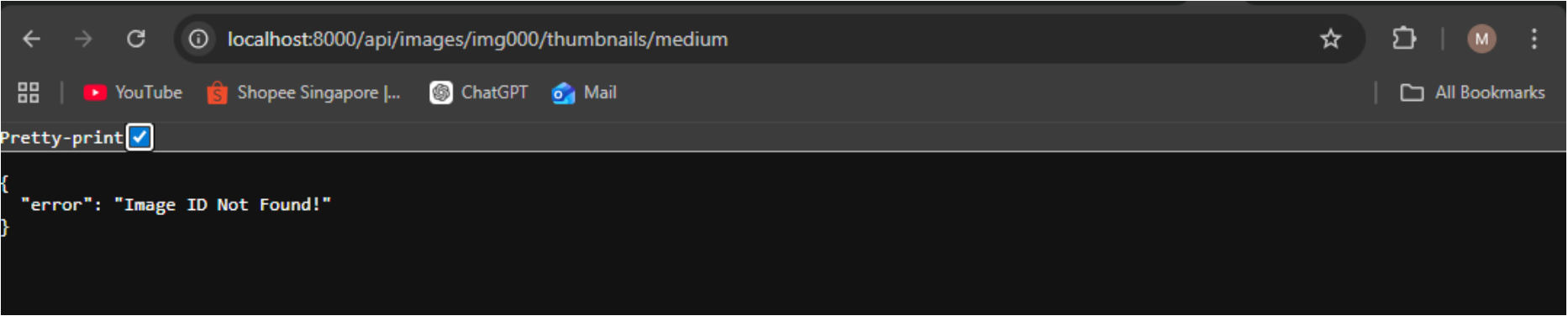
TC_009	Retrieve medium thumbnail successfully	http://localhost:8000/api/images/img932/thumbnails/medium	Image of cat_01.jpg but in medium size...	*Refer to Evidence	Pass
Evidence:					

localhost:8000/api/images/img932/thumbnails/medium

YouTube Shopee Singapore ChatGPT Mail

All Bookmarks



TC_010	Request for unsupported size (e.g. large)	http://localhost:8000/api/images/img932/thumbnails/large	Error message saying “Invalid Thumbnail Sizing...”	*Refer to Evidence	Pass
Evidence:					
					
TC_011	Thumbnail request for non-existent image_id	http://localhost:8000/api/images/img000/thumbnails/medium	Error message saying “Image ID Not Found”	*Refer to Evidence	Pass
Evidence:					
					

Caption Generation					
TC_012	Caption generation for valid image	curl.exe http://127.0.0.1:8000/api/images/img932	ai_caption should have a descriptive statement of the image...	*Refer to Evidence	Pass
Evidence: <pre> PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe http://127.0.0.1:8000/api/images/img932 {"status":"success","data":{"image_id":"img932","original_name":"cat_01.jpg","processed_at":"2026-02-23T11:55:09Z","metadata":{"width":270,"height":187,"format":"JPEG","size_bytes":7889},"exif_data":{"info":"No EXIF Data to extract"},"ai_caption":"a cat sitting on the ground next to a wall","thumbnails":{"small":"http://localhost:8000/api/images/img932/thumbnails/small","medium":"http://localhost:8000/api/images/img932/thumbnails/medium"}}, "error":null} PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> </pre>					
Response Schema & Error Handling					
TC_018	Successful response matches schema	curl.exe http://127.0.0.1:8000/api/images	* Showing the correct structure as per specified in the instructions	*Refer to Evidence	Pass
Evidence: <pre> PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> curl.exe http://127.0.0.1:8000/api/images [{"status":"success","data":{"image_id":"img932","original_name":"cat_01.jpg","processed_at":"2026-02-23T11:55:09Z","metadata":{"width":270,"height":187,"format":"JPEG","size_bytes":7889},"exif_data":{"info":"No EXIF Data to extract"},"ai_caption":"a cat sitting on the ground next to a wall","thumbnails":{"small":"http://localhost:8000/api/images/img932/thumbnails/small","medium":"http://localhost:8000/api/images/img932/thumbnails/medium"}}, "error":null}, {"status":"success","data":{"image_id":"img750","original_name":"cat_02.jpg","processed_at":"2026-02-23T11:56:38Z","metadata":{"width":277,"height":182,"format":"JPEG","size_bytes":8104},"exif_data":{"info":"No EXIF Data to extract"},"ai_caption":"a cat and a cat looking at each other","thumbnails":{"small":"http://localhost:8000/api/images/img750/thumbnails/small","medium":"http://localhost:8000/api/images/img750/thumbnails/medium"}}, "error":null}, {"status":"success","data":{"image_id":"img500","original_name":"hosue_01.jpg","processed_at":"2026-02-23T11:56:40Z","metadata":{"width":275,"height":183,"format":"JPEG","size_bytes":12607},"exif_data":{"info":"No EXIF Data to extract"},"ai_caption":"a large stone building next to a body of water","thumbnails":{"small":"http://localhost:8000/api/images/img500/thumbnails/small","medium":"http://localhost:8000/api/images/img500/thumbnails/medium"}}, "error":null}, {"status":"failed","data":{"image_id":"img029","original_name":"book_1.xlsx","processed_at":"","metadata":null,"exif_data":null,"ai_caption":"","thumbnails":null}, "error":"Invalid File Format!"}] PS C:\Users\muham\OneDrive\Desktop\HTX_API_Submission\Image_Processing_Pipeline> </pre>					

TC_019	Failure response matches schema	curl.exe http://127.0.0.1:8000/api/images/img029	* Showing the correct structure when encounter an error as per specified in the instructions	*Refer to Evidence	Pass
Evidence: 